Stanley Nelavala

Software Engineer: Payments, Java, Spring Boot, SQL, AWS
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Huntington, IN | Open to relocate anywhere

Summary

Software engineer who builds payment services that stay correct under retries, failures and load. Public Java and Spring Boot payments ledger (authorize, capture, void, refund, transfer) with idempotency keys, a transactional outbox, SLOs and a scripted failure lab, plus a C# card dispute service. At work: system design, secure APIs and SQL tuning for a multi-tenant platform. M.S. Computer Science (4.0 GPA). AWS Certified Solutions Architect, Associate.

Experience

Software Development Engineer Nov 2025 – Present
Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; numbers from dev and test) *Huntington, IN*

- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Replaced offset paging with keyset paging on the catalog device list: on a 328,000-device dev database, the query for row 150,000 fell from 285 ms to 0.20 ms. Pages never skip or repeat rows, tested with rows inserted, deleted and tied between pages.
- Fixed a regular expression denial-of-service bug in a shared validation package: an 80-character hostile input took about 138 s. The new linear pattern matched the old results on 60,000 random inputs.
- Standardized malware scanning of file uploads (AWS GuardDuty) across 9 app modules, and automated weekday start and stop of dev and acceptance servers (ECS, RDS): about 53 hours a week instead of 168.
- Built RxDesign, the hospital design service, backend and web (FastAPI, React, TypeScript), with 220+ offline API tests; set up separate iOS and Android release pipelines in GitHub Actions.

Adjunct Faculty (part-time) Jan 2026 – Aug 2026
The College of St. Scholastica *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

Graduate Research Assistant and Teaching Assistant Aug 2023 – Aug 2025
University of Minnesota Duluth *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR) that analyzed 1M+ TikTok videos from 300k+ authors for thesis research. Led labs for Python, NLP and Parallel Programming.

Payments Projects (public code, tests)

Payments Ledger Service | *Java 25, Spring Boot, PostgreSQL, Flyway, Kafka API, Prometheus, Grafana, k6* GitHub

- Double-entry ledger (authorize, capture, void, refund, transfer) with idempotency keys and a transactional outbox to a Kafka API (Redpanda); property tests and a k6 check that no money is created or lost.
- Application resiliency: scripted failure lab (Kafka down, Postgres restart, relay crash, poison event, latency injection) with balance checks after each, and a dead-letter topic with once-only redrive.
- Operational stability: three user-facing SLIs, 28-day error budgets, six burn-rate alerts tested with promtool and a Grafana SLO dashboard; rollback by image digest across an expand-and-contract Flyway migration; STRIDE review.

Ledger Ops Console and QA Suite | *React, TypeScript, Sass, Playwright, Cucumber, REST Assured, Selenium* GitHub

- React 19 ops console over read-only endpoints with one keyset paging contract: 32 Vitest and 13 Playwright tests, zero axe violations (WCAG 2.1 A and AA). Java QA suite: 41 black-box API tests (16 Cucumber scenarios, REST Assured and JSON-schema contract tests), 6 Selenium tests, 0 flaky.

Card Dispute Service | *C#, .NET 10, ASP.NET Core, EF Core, PostgreSQL, xUnit, CloudFormation* GitHub

- Dispute state machine with an append-only event table; concurrent decisions on one case produce exactly one winner (PostgreSQL xmin optimistic concurrency); idempotent create; tests written first (30 unit, 32 integration).
- Non-root image with a clean Trivy scan, Prometheus metrics and readiness checks; CloudFormation (ECS Fargate) linted and policy-checked, not deployed.

Skills

Languages: Java, SQL, Python, TypeScript, JavaScript, C#
Backend and data: Spring Boot, REST, idempotency, outbox, Kafka API, PostgreSQL, Flyway, FastAPI, ASP.NET Core, Redis
Resiliency and security: SLOs, burn-rate alerts, Prometheus, Grafana, failure labs, STRIDE, Trivy
Testing: JUnit 5, Testcontainers, Cucumber, REST Assured, Selenium, Playwright, Vitest, xUnit, pytest
Cloud, CI/CD, front end: AWS (ECS, RDS, S3, GuardDuty), GitHub Actions, Docker, Terraform; React, TypeScript, Sass
AI-assisted development: Claude Code, Codex, opencode, DeepSeek; outputs checked with tests, scans and review

Education and Certification

M.S. Computer Science, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Summer Research Fellowship 2024, 2025
B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 – 2023
AWS Certified Solutions Architect, Associate, Feb 2025 – Feb 2028 (verifiable on Credly)